

SUNIL GAUTAM

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5.8 years of experience specializing in React.js, Next.js, and TypeScript, delivering production-grade frontend applications for enterprise-scale platforms serving tens of thousands of users. Expertise in component architecture, design systems, real-time data interfaces (WebSockets, SSE), and performance optimization — driving 60–80% reductions in API calls and re-renders. Proven track record of owning complex frontend features end-to-end, driving scalable architecture, API integration, performance, accessibility (WCAG 2.2), testing, and production delivery. Experienced in legacy-to-modern migrations and cross-functional collaboration with backend, design, and QA teams in Agile/Scrum environments.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS/Sass, Python, C++

Frameworks & Libraries: React.js (Hooks, React 18/19), Next.js (App Router), Redux Toolkit, TanStack Query, Context API, React Router

UI, Styling & Design Systems: Tailwind CSS, Material UI, ShadCN UI, Ant Design, Bootstrap, Design Systems, Reusable Component Libraries

Architecture & Performance: Component-Driven Development, Micro-Frontends, Code Splitting, Lazy Loading, Memoization, Virtualization / Windowing, Core Web Vitals (LCP / CLS / INP), Lighthouse, PWA, Service Workers, Web Workers, SSR / SSG, Responsive Design, SEO, Accessibility (WCAG 2.2)

APIs & Real-Time: REST APIs, GraphQL, Apollo Client, WebSockets, Server-Sent Events (SSE), Polling, Real-Time / Async UX

Data Viz & Mapping: AG Grid, TanStack Table, Material React Table, Highcharts, Chart.js, Leaflet, OpenLayers, Google Maps

Auth & Security: OAuth 2.0 / OIDC, Keycloak, NextAuth, JWT, RBAC, ABAC

Testing: Jest, React Testing Library, Storybook, Playwright (E2E)

Build & Tools: Vite, Webpack, Babel, Git, GitLab CI/CD, Bitbucket, Docker, ESLint, Prettier, Husky, npm / pnpm / Yarn, GitHub Copilot (AI-Augmented Development), Anthropic API / AI Integration

Backend & Data: Node.js, Express.js, MongoDB, Neo4j, Kafka, .NET Core (integration), REST / GraphQL API design

PROFESSIONAL EXPERIENCE

Senior Software Engineer • Dot AI

Feb 2024 – Present

AI-powered logistics & asset-intelligence SaaS platform (Dot Oracle) integrated with IIoT hardware for real-time asset monitoring and tracking.

- Reduced JS bundle size ~30% by analyzing bundle composition, redesigning client-side loading boundaries, introducing **route-level splitting / dynamic imports**, and eliminating unnecessary dependencies from critical paths.
- Improved initial load performance ~35% for data-intensive workflows by optimizing the **critical rendering path**, deferring non-critical UI, optimizing image/font delivery, and tuning prefetching/caching strategies; validated via **Lighthouse**.
- Sole frontend owner of **13 of 17 core modules**, built end-to-end on a scalable, **component-driven React + TypeScript** architecture — from design collaboration to production deployment.
- Re-engineered **real-time map tracking** to render **10,000+ live asset markers** with smooth pan/zoom using Leaflet clustering, viewport culling, and batched marker updates.
- Cut wasted re-renders and main-thread work through **memoization, list virtualization / windowing**, and render optimization — improving **interaction latency (INP)** and scroll smoothness on high-density views.
- Led a **zero-downtime migration** from legacy UI to a modern design system using a **Next.js App Router** hybrid-routing strategy that let old and new UIs coexist for a gradual, non-disruptive rollout.
- Improved **accessibility (WCAG 2.2)** across the component library by implementing **keyboard navigation, focus management, and semantic HTML5** to deliver inclusive, standards-compliant UI.
- Implemented **enterprise authorization (Keycloak, OAuth 2.0 / OIDC, RBAC/ABAC)** and established **Jest / RTL testing** standards; partnered with Product, UX, backend, and DevOps on REST/GraphQL contracts and CI/CD.
- Mentored **junior engineers** through structured **code reviews**, technical design discussions, and enforcement of frontend engineering standards, improving code quality and cross-team consistency.
- Integrated the **Anthropic Claude Code API** into the development workflow to auto-identify and fix related test cases on source changes, reducing manual testing overhead across the team.

- Led enterprise analytics dashboards and high-density data-visualization workflows (React.js, Redux, AG Grid, Highcharts), delivering production virtualized grids and charts with improved rendering performance.
- Built reusable, component-driven frontend architecture (React.js, TypeScript, SCSS, Material UI) to improve maintainability and UI consistency across the design system.
- Delivered full-stack beyond the UI — designed and built **REST APIs in .NET Core**, modeled and queried highly connected data in a **Neo4j graph database**, and integrated **Apache Kafka** event streams to drive real-time data pipelines feeding the dashboards.
- Managed framework upgrades, dependency modernization, and third-party integrations with minimal production impact; collaborated across sprint planning, backlog refinement, and estimations in Agile cycles.

Software Engineer • Daffodil Software

Jan 2021 – Sep 2021

- Built and shipped production features for scalable React.js web applications, developing reusable, responsive UI components and asynchronous data-fetching workflows that improved code reuse and UI consistency.
- Delivered full-stack functionality end-to-end across **React.js, Node.js, Express.js, and MongoDB** — designing and integrating REST APIs, modeling data, and wiring backend services to the UI.
- Restructured and optimized Express API endpoints for better query efficiency and faster response times, and rapidly diagnosed and resolved critical production issues to keep releases stable.

KEY PROJECTS

Dot Oracle — AI Asset-Intelligence SaaS Platform • Dot AI

Enterprise SaaS that ingests live IIoT telemetry to monitor, track, and analyze physical assets in real time across logistics and industrial operations — sole frontend owner of 13 of 17 core modules.

- Architected and rebuilt the platform on a scalable **Next.js 14 (App Router) + React + TypeScript + Material UI v5** foundation, establishing a shared design system and reusable component library adopted across every module for consistent UX and faster feature delivery.
- Engineered the **real-time operational map engine** — Leaflet marker clustering, viewport culling, and batched marker updates streamed over WebSocket/SSE — to render tens of thousands of live IIoT assets with smooth pan/zoom and near-instant status and alert updates.
- Built high-density enterprise workflows — virtualized data grids, multi-step wizard flows, bulk operations, and live operational dashboards — tuned for large datasets with memoization, windowing, and request caching to keep interactions responsive under heavy load.
- Drove a **zero-downtime legacy-to-modern migration** via a hybrid App Router routing strategy, and hardened the platform with Keycloak authorization (RBAC/ABAC, OAuth 2.0 / OIDC), WCAG 2.2 accessibility, and Jest/RTL-tested components for long-term maintainability.

CoBoard — Real-Time Multiplayer Whiteboard • Personal Project [Live Demo](#)

Real-time, infinite-canvas multiplayer whiteboard — create a board, share the link, and draw together live with no signup.

- Built a full whiteboard engine from scratch (freehand drawing, shapes, arrow binding, bound text labels, selection/transform, infinite pan-zoom) with **React, TypeScript, and react-konva**.
- Engineered an operation-based real-time sync protocol on Supabase Realtime with last-write-wins conflict resolution and message-budget-aware broadcasting (throttled cursor/drag updates, payload caps).
- Added state persistence and recovery (debounced Postgres snapshots, hydration replay, auto-reconnect with exponential backoff) and shipped an accessible, mobile-capable UI on Vercel at zero infra cost.

EDUCATION

B.Tech in Computer Science and Engineering — Chandigarh University

2017 – 2021 • CGPA 7.69

ACHIEVEMENTS & RECOGNITION

- **Zero-Downtime Platform Migration — Dot AI:** architected a phased migration letting legacy and modern UI systems coexist, with no disruption to active enterprise customers.
- **High-Impact Contribution Recognition — Dot AI:** recognized for technical ownership, decisive problem-solving, and delivery, later advancing into frontend leadership.
- **“Pat on the Back” Award (×2) — Hexagon:** delivering critical features on time with strong ownership and execution.
- **Reward & Recognition — Daffodil Software:** rapid issue resolution, product-quality improvements, and timely delivery of critical fixes.